

An empirical study of the relationship between inclusive leadership and business model innovation

Yangchun Fang

*Global Institute for Zhejiang Merchants Development,
School of Management, Zhejiang University of Technology, Hangzhou, China, and
Xinxing Dai and Xudong Zhang
School of Management, Zhejiang University of Technology,
Hangzhou, China*

Abstract

Purpose – To better understand factors that lead to business model innovation (BMI) in organizations, this study argues that inclusive leadership is the primary source that motivates employee engagement in boundary-spanning activities, which fosters BMI by generating and integrating employee knowledge through boundary-spanning exploration.

Design/methodology/approach – This paper analyzes the relationships between inclusive leadership, boundary-spanning exploration and BMI to discover how they influence each other. This is achieved by a survey of more than 30 enterprises in Zhejiang Province, China. The survey data are analyzed using SPSS.

Findings – Inclusive leaders who display characteristics of openness, accessibility and availability have a positive effect on BMI and boundary-spanning exploration; openness and accessibility have more significant positive effect on BMI and boundary-spanning exploration.

Research limitations/implications – Because this study covers only in Zhejiang Province, in China, the findings may not be easily generalizable to other contexts. Other factors, such as organizational structure and contextual characteristics, could also be added as antecedents in a future investigation. Moreover, the scales could be revised more suitable for China in the future.

Practical implications – The empirical findings can be used as guidelines for firms seeking to take advantage of inclusive leadership and to help firms motivate their employees to engage in boundary-spanning exploration to acquire, generate and integrate knowledge from diverse sources to promote BMI.

Originality/value – This study provides insights that will be of value by adding to the theoretical foundation of our understanding of the antecedents to firms' BMI. This study argues that employees' boundary-spanning exploration is important to BMI as well. Inclusive leadership is crucial to stimulate boundary-spanning exploration by employees. Therefore, how to encourage inclusive leadership in firms is well worth examining.

Keywords Inclusive leadership, Boundary-spanning exploration, Business model innovation

Paper type Research paper

Introduction

Many large firms such as Google and Alibaba succeeded because of technology and for their adoption of business model innovation (BMI). BMI involves repositioning value propositions, redesigning profit formulas and identifying key resources (Johnson *et al.*, 2008), allowing firms to reconfigure their value creation mechanisms. Although scholars have widely confirmed the advantages of BMI, understanding of its antecedents remains fragmented. Previous studies of BMI have discussed antecedent factors such as strategy (Teece, 2010), technology (Chesbrough, 2010) and organizational structure (Amit and Zott, 2001). However,

Funding: This research was supported by the National Social Science Fund Project of China (Grant No. 20BGL143), the Social Science Project of Zhejiang University of Technology–School Pre-research Fund Project (Grant No. SKY-ZX-20200121).



BMI's success cannot be separated from a firm's employees. As a typical case, in Alibaba's journey from providing the online C2C trading platform Taobao to diversification into new product areas such as Ant Financial Services, successful BMI not only involved technological transformation but also innovative ideas generated by individuals in the organization. Alibaba's CEO, Daniel Zhang, noted in 2019 that individual diversity, creativity and innovation are crucial to BMI. However, integrating diverse employees into the workplace is challenging because of their different preferences and backgrounds (Warner and Zhu, 2010). China is a leading emerging market, and the rapidly increasing diversity of China's workforce has necessitated acknowledging the importance of inclusive leadership (He *et al.*, 2007; Peng *et al.*, 2008). Successful inclusive leadership allows companies to leverage and include various talented employees while simultaneously contributing to employee job satisfaction and well-being (Jiang and Chen, 2018; Tang and Naumann, 2016). Inclusive leadership can be an important issue for a firm's BMI.

Inclusive leadership as a type of relational leadership has received recognition from researchers and practitioners as a means of coping with the Chinese workforce's increasing diversity. Inclusive leadership welcomes individuals from various demographic and cultural backgrounds (Mor Barak, 2000), supports their involvement and appreciates their participation (Nishii, 2013). Scholars define inclusive leadership as having the specific characteristics of openness, accessibility and availability (Carmeli *et al.*, 2010) in leaders' interactions with followers. That is, leaders allow employees to share their views and inputs by being open, available and accessible, thus helping to shape the belief among employees that "their voices are genuinely valued". However, despite its importance, understanding the specific characteristics or leadership behaviors that lead to BMI is lacking. Little empirical research has been conducted on the relationship between inclusive leadership and BMI. Therefore, it needs more studies to understand the ways and mechanisms by which inclusive leaders cultivate the right conditions and enhance innovation by employees to promote BMI and of the moderating or mediating variables related to these results.

One major obstacle to BMI is organizational inertia (Amit and Zott, 2015), of which cognitive inertia and resource and path dependence are the most prominent types. To break through organizational inertia, enterprises must encourage employees to go beyond organizational boundaries and dependence on inherent resources by engaging in boundary-spanning exploration. This facilitates firms in increasing the possibility of obtaining new knowledge by accessing various knowledge sources (Leiponen and Helfat, 2010). However, boundary-spanning exploration has some obstacles, including role overload, assimilation problems and self-efficacy issues. Thus, how to stimulate diverse employees to engage in boundary-spanning exploration is worthy of further research because it can contribute to a company's BMI. This study examines whether inclusive leadership is conducive to BMI by investigating its influence on employee willingness to make efforts to engage in boundary-spanning exploration, thus leading to creative ideas.

The purpose of this study is to explore the core question of inclusive leadership's positive role in BMI, and how inclusive leadership affects BMI conceptually and empirically. This is accomplished by proposing and testing a mediation model of the relationship between inclusive leadership and BMI, which incorporates boundary-spanning exploration as a mediator. This study aims to extend the existing research literature in several ways. First, it departs from the common scholarly practice of treating technological or organizational structures as the dominant antecedents of BMI and focuses instead on inclusive leadership's predictive power. This study contributes to the development and refinement of the extant literature on the relationship between leadership and BMI. Second, by drawing on open innovation theory, this study opens the black box that is the relationship between inclusive leadership and BMI from a boundary-spanning exploration perspective. The focus on the role of employees in developing boundary-spanning exploration that facilitates individual

innovation responds to the calls for more innovation in BMI. Third, by integrating social exchange theory and open innovation theory, this study views individual behaviors as a significant factor in the model and examines the employee contributions to the organization. It addresses the important question of what matters to employees and enriches existing BMI research, which has largely neglected the analysis of individuals in organizations.

Literature review and hypotheses development

Inclusive leadership and business model innovation

Amit and Zott (2001) defined business models as containing an organization's transaction content, structure and governance; they argued that different designs of these three make business models display different design themes such as efficiency, novelty, lock-in and complementarity, and that this process leads to the business model's innovation. BMI is considered a critical antecedent of firm performance because it involves adopting new ways of conducting economic exchanges (Zott and Amit, 2007). According to Teece (2010), BMI simply involves discovering new or totally different modes of value proposition, capture and creation. BMI as an effective vehicle is achieved by adjustments and improvements to key elements in firms, including creating, implementing and validating activity (Massa and Tucci, 2013). Nonetheless, companies often find themselves struggling to overcome barriers to facilitate BMI; for example, some scholars investigating BMI have studied its relationship with technological innovation (Khanagha and Volberda, 2014), internal resources and capabilities (Mezger *et al.*, 2016), the senior leadership (Baden *et al.*, 2013) and organizational structure adjustment (Mezger, 2014); however, few papers about BMI have emphasized the role of employees. Employees' generation and implementation of novel ideas, processes or procedures within a company, which do not simply occur but require key resources and encouragement, are favorable to BMI (Martins and Terblanche, 2003). Autonomy for idea generation and creative problem solving is also necessary (Zhou, 1998). Therefore, support and encouragement from leaders for employees to take the initiative and explore innovative ideas are likely to affect the actual innovation of the business model.

Leadership is particular important to influence employees' creativity and innovation in organizations (Mumford and Hunter, 2005). Leaders can energize and motivate employees to engage in creative task (Atwater and Carmeli, 2009), provide necessary resource for creative endeavor (Reiter-Palmon and Illies, 2004) and support creative behaviors by providing high-quality relationship between leaders and employees (Carmeli *et al.*, 2010). Leader's support is always related to motivation to creative behaviors. However, with an increasingly diverse workforce, companies are struggling to retain high-quality employees and stimulate their innovation. It needs more studies to understand which leader behaviors can cultivate conditions for diversity employees to enhance creativity. Recently, with the study on diversity, scholars focus on inclusive leadership. The concept of inclusive leadership was first put forward by Nembhard and Edmondson (2006). They named it leader inclusiveness, which means leaders invite and appreciate inputs from followers. Afterwards, based on leader-member exchange theory, Hollander (2009) systematically defined that inclusive leadership is follower-centered, respect employees and understand their needs. Learning from Nembhard and Edmondson' views and taking over Hollander's ideas, Carmeli *et al.* (2010) extended them appropriately. They defined inclusive leaders in terms of leaders who exhibit openness, accessibility and availability, which is widely applied and recognized in academic circles. Carmeli *et al.* (2010) considered that inclusive leadership attends to the relationship between leaders and employees and focuses on the support and assistance to employees, pay close attention to employees "needs to support their suggestions, as well as encourage employees" involvement in organization activity. Thus, this paper uses this construct of inclusive leadership presented by Carmeli *et al.* (2010) for empirical analysis.

Based on social exchange theory (Blau, 1964), a high-quality exchange relationship between leaders and employees generates effective interactions between them, therefore enabling employees to increase their intrinsic motivation and cultivate energy to engage positively in work. Thus, if leaders treat employees in a positive and friendly way, employees will respond in an equivalent or even more positive manner. Inclusive leadership view their own role as support and assistance, meaning it is essential to offer employees necessary resources, independence and direction to perform their work (Carmeli *et al.*, 2010; Hollander, 2009). Thus, when firms expand their business or engage in BMI, inclusive leader is available to provide relational support and offer followers access to resources necessary to create original ideas that will lead to increased creative production, procedural innovation or ways of participating in transactions, which are important elements of BMI. In addition, inclusive leadership helps develop strong emotional links and interpersonal relationships within an organization by practicing socio-emotional support behaviors effectively (Hollander, 2009). This establishes an inclusive working atmosphere and a warm, open and supportive culture by conducting inclusive activities that encourage employees to think in terms of cooperation, resource attainment and productive creativity (Carmeli *et al.*, 2010). Positive peer relationships, emotional attachment and high-quality exchange are important innovation-enhancing forces enabling members to exhibit their talent in an organization (Hunter *et al.*, 2007).

In addition, inclusive leaders enhance the inherent motivation of employees by showing concern for them and considering their interests, expectations and feelings and by being available and accessible as employees accomplish their tasks (Carmeli *et al.*, 2010; Hollander, 2009). Therefore, inclusive leaders help employees take the initiative and explore innovative solutions likely to influence the degree of boundary-spanning exploration, which enables firms to fit better within varying environments, benefits firms by acquiring heterogeneous resources, complements firms' internal resources and eventually creates more opportunities for firms to conduct BMI (Zhang and Hu, 2017). In addition, inclusive leaders recognize and respect employee differences and encourage them to try different ways of finding solutions without worrying about the negative consequences that might result from such behavior (Carmeli *et al.*, 2010; Hollander, 2009). By sending a strong signal that various alternative ideas are welcome, inclusive leaders make employees feel recognized and appreciated for their uniqueness and differences (Hollander, 2009), thus reducing the pressure to maintain consensus and allowing employees to generate, promote and realize new ideas for BMI.

H1. Inclusive leadership positively correlates with BMI.

Inclusive leadership and boundary-spanning activity

Boundary-spanning exploration is relatively vaguely defined in the scholarly literature. However, the concept of boundary-spanning activity has mostly been studied from the perspectives of evolutionary economics and the resource-based view. In evolutionary theory, a primary assumption is "local search," meaning the search behavior of any firm is in the neighborhood of its current knowledge (Stuart and Podolny, 1996). From the resource-based view, the previous literature stresses that an enterprise's sustainable competitive advantage depends on its ability to move beyond local search and rearrange its expertise resources. Such an ability has been described as a "combinative capability" (Kogut and Zander, 1992), "architectural competence" (Henderson and Cockburn, 1994) and "dynamic capability" (Teece *et al.*, 1997). Empirical evidence suggests that firms that engage only in local search and focus on technological domains closely related to their own do not achieve outcomes that are systematically distinguished from those of competitors.

Many scholars argue that boundary-spanning activity fosters innovation (Leiponen and Helfat, 2010). However, less attention has been given to understanding potential obstacles.

The first is role overload because the boundary-spanning activity can compete with internal activity for attention, thus leading to internal pressure (Katz and Kahn, 1978). Individuals are unlikely to engage in these external activities unless they consider themselves part of the internal organization. The second obstacle is boundary-spanning self-efficacy: without self-efficacy, people do not have the necessary confidence to successfully perform boundary-spanning behaviors (Bandura, 1997) and successfully manage relationships with important external parties. In addition, boundary-spanning activity is accomplished only by those individuals who are well connected to both external and internal parties because they are able to disseminate new knowledge to their locally oriented colleagues.

How, though, can this phenomenon be managed? This study proposes that inclusive leaders can facilitate individual boundary-spanning behaviors. First, inclusive leaders' intentions of ensuring justice and equity, sharing decision-making and maintaining emotional identification among employees can increase individuals' sense of group identity, thus enhancing the followers' sense of being members of a group and demonstrating that they belong to the organization (Easterbrook and Vignoles, 2013; Liu *et al.*, 2012; Tyler *et al.*, 1996). The sense of belonging and emotional attachment within an organization decrease the risk posed to individuals' internal relationships by introducing new ideas from outside parties. Accepting an employee as an internal member (Stamper and Masterson, 2002) relieves the problem of role overload and internal pressure when employees perform boundary-spanning activities. Second, inclusive leaders improve individuals' sense of self-efficacy. When inclusive leaders show respect for and engage in fair treatment of diverse employees, attend to their needs, recognize employees' unique characteristics, and bear the trial-and-error cost, leaders capture the employees' goodwill and trust, thus forming a relationship of mutual trust and reciprocity between employees and leaders (Fang, 2014). In such a positive relationship, employees can establish a positive psychological state and improve the confidence and level of optimism of employees regarding future tasks and allows employees to become confident that they can overcome difficulties to achieve their work goals, thus raising their self-efficacy (Snyder, 2005). Moreover, inclusive leaders emphasize individuals' uniqueness (Nishii, 2013), which contributes to employees perceiving that they are valued for their different qualities and contributions. Inclusive leaders consider information and perspectives that employees have explored in outside regions, thus creating the perception that employees are valued for what they create and learn from internal and external sources. By encouraging and recognizing different contributions and divergent viewpoints, employees come to feel that they can safely present new ideas and take risks involved with developing ideas that basically defy the norm (Kearney and Gebert, 2009). Therefore, inclusive leaders enable employees to disseminate new knowledge to their locally oriented colleagues.

H2. Inclusive leadership positively correlates with boundary-spanning exploration.

Boundary-spanning exploration and business model innovation

In high-tech industries, a firm's success depends on constant innovation, which is related to the ability to acquire, generate and integrate knowledge from both external and internal sources (Leonard-Barton, 1995). Therefore, it is crucial for enterprises to seek various kinds of knowledge from diverse sources. Laursen and Salte (2006) observed that boundary-spanning exploration guides enterprises to search for new ideas and knowledge; in the process of BMI, a greater breadth of partner enterprises with which to interact leads to better innovation outcomes. Ruef (2002) found that business teams embedded in heterogeneous working networks were more likely to discover the innovative ideas essential for BMI than those in homogenous ones. Diversity of expertise in scientific and technical organizations increases

innovation results and contributes to higher rates of innovation breakthroughs (Fleming *et al.*, 2007).

From the perspective of open innovation theory, in responding to the rapidly changing economic environment, it is highly important for companies to break through old closed organizational and technological borders and accept more innovative ideas from the external environment (Chesbrough, 2003a, b, c), which reduces organizational inertia and maintains adaptability. The idea that acquiring a large range of new knowledge accelerates product innovation and technological process improvement has been widely confirmed (Ahuja and Lampert, 2001). The larger the search scope is, the more knowledge is acquired and the harder it is to imitate the integrated capabilities (Smith and Tushman, 2005), which is the key link for enterprises to build core competitiveness and promote BMI. Therefore, this paper argues that boundary-spanning exploration has a positive effect on BMI.

H3. Boundary-spanning exploration plays an intermediating role between inclusive leadership and BMI.

Methodology

Sample procedure and data collection

The sample and data for this study were collected from high-tech enterprises in Zhejiang Province, China, that operates in the fields of telecommunications, manufacturing and e-commerce. With the rapid development of and fierce competition among high-tech enterprises, BMI is an urgent priority, making it is easier for us to observe the relationship between BMI and inclusive leadership. The questionnaire used in this study was originally composed in English and then translated into Chinese by a double back-translation process. To ensure content validity, the questionnaire was discussed with firm managers and professors, who were asked for their suggestions. Based on their feedback, we modified the questionnaire to correct discrepancies between the Chinese and English versions. We then conducted a preliminary test to ensure the exact understanding of each survey item by 100 respondents from five enterprises. The survey items were expressed with neutral words; we also needed to inform respondents of the academic purpose of the study and assure them of the confidentiality of their responses.

We then selected 30 companies to participate in our survey, sending each 20 questionnaires. We phoned the leaders listed in our sample to ask for their agreement to take part in the survey. Our respondents were primarily top managers or leaders of departments. If the directors were not available, we asked heads of key departments to fill in the questionnaire individually or together. It should be noted that only those who were knowledgeable concerning their firms' BMI were qualified to take part in the interviews. Among our respondents, over 85% had worked in the company for more than five years, with an average of 7.5 years of business tenure. This length of service ensured that they were qualified interviewees. In addition, to eliminate common method variance, the BMI scale and boundary-spanning exploration scale items were filled out by department leaders or CEOs, and that on inclusive leadership style was filled out by department employees.

Measure and validation

All survey items were measured on a five-point Likert scale where 1 was "strongly disagree," and 5 was "strongly agree."

Independent variables. This study used the measurement scale developed by Carmeli *et al.* (2010), who argued that inclusive leadership has three dimensions: openness, accessibility and availability. These dimensions were tested for by including items such as "the boss is

willing to listen to new plans or ideas” and “the boss is concerned about new opportunities to improve work processes.”

Dependent variable. As mentioned above in the theoretical overview, there is still some debate as to how to define BMI. Of those scholars who have approached this question, the work of Zott and Amit is well regarded for having made extensive use of quantitative measurement to study BMI. This study used the measurement scale developed by Zott *et al.* (2007), with items including “providing customers with products or services with increasing value” and “constantly acquiring a large number of new and diverse customers.”

Mediating variable. The dimensions of boundary-spanning exploration vary with the research emphasis; it is often divided into the dimensions of knowledge distance and knowledge type. The types of knowledge involved can be divided into technical knowledge and market knowledge (Sofica and Grimpe, 2010). This paper measures boundary-spanning exploration of technical knowledge and market knowledge using the scale developed by Sidhu *et al.* (2007), which has been widely recognized by scholars.

Control variables. To reduce the potential impact of endogenous problems in the research and enhance the robustness of the results, this paper takes enterprise size, age and industry as control variables.

The study implemented several methods of analysis to test for reliability and validity. It used Cronbach’s alpha and composite reliability (CR) to estimate the internal consistency reliability. As Table 1 shows, the CR scores were all above 0.7, meaning that they were adequate; then, to test the convergent validity, exploratory factor analysis (EFA) and confirmatory factor analysis (CFA) were used. All factor loadings were above 0.7, and the average variance extracted (AVE) for each scale was greater than 0.5. Finally, for the discriminant validity, as shown in Table 2, every extracted root of AVE was greater than the figures in the lateral and column orders. Therefore, the scales are confirmed to have good convergence and discriminant validity.

In addition, this study used the variance inflation factor (VIF) to examine the possibility of multicollinearity. All VIF values were around 1.5, far below the benchmark of 10, so no significant multicollinearity problems were present in the data. Furthermore, this study used different sets of respondents in the same enterprise to complete each type of questionnaire to

Table 1.
Reliability analysis and
convergent validity

Variables	Cranach’s α	CR	AVE
Inclusive leadership	0.911	0.911	0.777
Business model innovation	0.896	0.904	0.760
Boundary-spanning exploration	0.829	0.824	0.541

Table 2.
Discrimination validity

Variable	1	2	3	4	5	6	8
IL1	<i>0.881</i>						
IL2	0.539**	<i>0.925</i>					
IL3	0.549**	0.530**	<i>0.836</i>				
BMI1	0.498**	0.516**	0.435**	<i>0.904</i>			
BMI2	0.406**	0.487**	0.347**	0.506**	<i>0.941</i>		
BMI3	0.390**	0.415**	0.379**	0.437**	0.530**	<i>0.954</i>	
BS1	0.515**	0.490**	0.407**	0.488**	0.382**	0.337**	<i>0.958</i>
BS2	0.478**	0.442**	0.442**	0.415**	0.350**	0.354**	0.423**

Note(s): (1) Significant at ** $p < 0.05$; (2) The italic number on the main diagonal is the square root of AVE

avoid the problem of common method variance, which can be confirmed with Harman's single-factor test. We put all variables into an EFA procedure with the principal axis factoring method (Podsakoff *et al.*, 2003). For the unrotated factor part, eight factors emerged with eigenvalues exceeding 1.0, with the first one explaining 38% of the variance. Thus, no single factor was dominant.

Finding

Descriptive statistics and correlation test

The sample had 180 valid questionnaires, collected with an effective response rate of 60% (180 out of 300). The average firm age was 29.5 years, and the average firm size was 1,000 employees. There were 53.11% private companies, 14.52% state-owned companies, 12.03% joint ventures and others such as foreign-owned companies and village firms. As for the industry, 31.54% of companies were from the technology industry, 26.5% from the manufacturing industry and the others were from industries such as wholesale and retail and culture and communication. Approximately 41.08% of the respondents were primarily administrative managers, 25.31% were from technology departments, and others were from marketing and production departments. More than 67% of respondents were well-educated, with 50% holding a bachelor's degree.

From the Pearson correlation in Table 2, it can be seen that BMI has a significant positive relationship with inclusive leadership, which means that it is possible to test for using hierarchical regression analysis. In model 1, the relationship between inclusive leadership and boundary-spanning exploration was tested; the control variables and independent variables were put into regression, which indicated that the *R*-squared was much greater than only the control variables in the model, from 0.067 to 0.447, meaning that inclusive leadership has a significantly positive correlation with boundary-spanning exploration, and, furthermore, that the dimensions of openness and accessibility have greater effects on boundary-spanning exploration. In model 2, boundary-spanning exploration was used as the independent variable and BMI as the dependent variable; it can be seen the *R*-squared is 0.346, which explained the 34.6 variance in BMI well, and the effective values are 0.313 and 0.296, which predict BMI better. Meanwhile, model 3 used inclusive leadership as the independent variable and BMI as the dependent variable; as predicted, they have a better fit relationship, with effective values of 0.247 (openness), 0.322 (accessibility), 0.138 (availability) and the *R*-squared is 0.425, which means it accounts for 42.5% variance in BMI. Therefore, the findings support Hypotheses 1 and 2.

For examining the mediating effect of boundary-spanning, this study adopted Mplus 7.4 to analyze the effect of the mediating role of boundary-spanning exploration. Table 3 shows that the fit indices (CMIN/DF, RMSEA, CFI, TLI, and SRMR) of the mediating role model are in accordance with the standard values, meaning that this model can be accepted. As Table 4 shows, this study generated five paths to predict the mediating effect. This showed that boundary-spanning exploration plays a good mediating role in BMI and inclusive leadership under a 95% confidence interval. Therefore, Hypothesis 3 is established.

Indicator	CMIN/DF	RMSEA	CFI	TLI	SRMR
Value	1.177	0.031	0.985	0.982	0.043
Standard	<3	<0.05	>0.9	>0.9	<0.1

Table 3.
Model fit analysis

Table 4.
 Mediating effect
 analysis

Discussion and implications

Discussion

Although there have been quite a few studies involving leadership and BMI, little research has focused on how to develop BMI in a globally diverse and inclusive environment. However, the idea that inclusive leadership is helpful in managing diversity and stimulating diverse employees to create new knowledge to facilitate BMI has recently begun to gain traction. Innovative talent has the ability to come up with new views and systems by communicating with or exploring diverse resources externally and internally, all of which lead to progress in BMI. Based on open innovation theory and the resource-based view, diverse resources and new knowledge are considered part of competitive advantage and are seen as necessary to break through organizational inertia and path dependence. Boundary-spanning exploration is an effective way to expand the scope of an organization and acquire heterogeneous resources. As practitioners, the employees are the driving factor in carrying out strategies and tasks, meaning that the way to stimulate employees' potential capabilities is worth exploring in depth. This study demonstrated that inclusive leadership has a significant positive effect on employees' boundary-spanning exploration and are able to

95% confidence interval					
The path	Estimate	S.E.	Est./S.E.	The lower	The higher
IL1-BS1-BMI	0.113	0.034	3.585	0.051	0.184
IL1-BS2-BMI	0.096	0.026	3.303	0.050	0.153
IL2-BS1-BMI	0.099	0.031	3.423	0.037	0.162
IL2-BS2-BMI	0.083	0.025	3.223	0.044	0.138
IL3-BS1-BMI	0.109	0.027	3.842	0.062	0.169
IL3-BS2-BMI	0.104	0.025	3.574	0.062	0.163

Table 5.
 Regression analysis

Variable	Business-spanning exploration		Model 2		Business model innovation	
	Model 1				Model 3	
<i>Controls</i>						
Firm age	−0.120	−0.034	−0.077	−0.150	−0.077	−0.007
Firm nature	0.141	0.021	0.278**	0.204**	0.278	0.157**
Industry	0.107	0.049	0.057	0.004	0.057	0.004
Firm size	−0.106	−0.054	−0.050	0.008	−0.050	0.003
<i>Independent</i>						
Openness		0.344**				0.247**
Accessibility		0.263**				0.322**
Availability		0.157**				0.138**
<i>Mediating variable</i>						
Boundary-spanning exploration 1				0.313**		
Boundary-spanning exploration 2				0.296**		
<i>Model statistics</i>						
R ²	0.067	0.447	0.100	0.346	0.100	0.425
Adjusted R ²	0.045	0.424	0.080	0.323	0.080	0.401
ΔR ²	0.067	0.380	0.100	0.246	0.100	0.324
F-value	3.128**	19.851**	4.872**	15.246**	4.872**	18.131**
Note(s): (1) Significant at ** $p < 0.05$						

Note(s): (1) Significant at ***p* < 0.05

motivate and support employees and inspire them to communicate with the external environment and explore the crossover of ideas.

The findings revealed that both inclusive leadership and boundary-spanning exploration are positively related to BMI. As shown in [Table 5](#), although theoretically inclusive leaders have a positive influence on boundary-spanning exploration and BMI, the two dimensions of openness and accessibility of inclusive leadership have a greater impact than the dimension of availability. Availability alone fails to serve as a distinct advantage for BMI. This can be explained by leaders with openness being likely to motivate and provide opportunities to encourage employees to engage in the exploitation and identification of new knowledge and markets. In addition, in terms of inclusive leadership as a specific form of relational leadership, accessibility represents a close relationship between employees and managers who have a strong desire to enable employees to solve problems they encounter during exploration and bringing in crucial information, knowledge and networking opportunities. These results collectively point to the importance of the role played by inclusive leaders in influencing BMI.

An additional discussion may be warranted on the effects of boundary-spanning exploration. BMI drives enterprises to search for innovative ideas internally and externally. According to open innovation theory, enterprises applying open innovation can broaden their BMI and promote the use of external knowledge, such that innovative internal ideas not only flow outward but also that external technology or ideas flow inward. However, BMI does not simply involve looking for new technologies externally but also requires adjustments by the company. The problem is that diverse resources may not be a good fit with the enterprise system or development situation. Employees need to use a variety of sources to explore effective technologies or knowledge to stimulate BMI. Thus, how to recognize and value physical and cultural differences in a meaningful way is important for employees. Companies must develop optimal business models to use technologies or knowledge effectively to compete with other companies. Moreover, on the firm level, there are diminishing returns to external search breadth ([Laursen and Salter, 2006](#); [Leiponen and Helfat, 2011](#)). Paying too much attention to searching for different external knowledge sources can, to some extent, be detrimental ([Koput, 1997](#)) because firms have a limited absorptive capacity ([Cohen and Levinthal, 1990](#)). Excessive external searching leads to wasted resources when firms gather more information than they can use. However, our study of the effects of boundary-spanning exploration, from an empirical research perspective, is limited to preventing possible trouble and is less sensitive to the particular needs of certain enterprises.

Implications

Overall, these results have considerable implications for both theory and practice. Although research on BMI has attracted a great deal of attention ([Chesbrough, 2010](#)), there is still a lack of clarity in defining the conceptual framework for the business model, and the antecedents of BMI are still controversial; therefore, by proposing a valuable leadership-based explanation for the business model, this study provides additional insight into the current debate on the antecedents of BMI and provides a new view on the definition of BMI.

As for management practice, novel business models involve exploring opportunities in the market ([Johnson et al., 2008](#)), meaning that leaders need to stimulate and attract employees to engage in boundary-spanning exploration to identify and shape such opportunities. While, in face of diversity in workforce, firms should adopt the inclusion management. Diversity is a core feature of inclusion, instead of avoiding the diversity, organization should manage the diversity, foster an inclusion culture to maintain and leverage diversity ([Tang et al., 2017](#)). Leaders as an important factor in organization should learn the inclusive leadership behaviors to manage the diversity. However, different individuals have their own

management features; enterprises should at first build an inclusive culture, treat leaders in an inclusive way, recognize their value and encourage them to show their talent. Firms could provide opportunities for leaders of different apartments to learn the concept and feature of inclusive leadership and encourage them to acquire and apply the inclusive leadership behaviors in management process. Under the inclusive culture, leaders will perceive the inclusion culture and apply inclusion into management. BMI benefits from a leader's openness, accessibility and availability by encouraging employees to gain access to knowledge, important resources and networking opportunities. Moreover, some scholars have raised doubts about the limits of inclusive leadership, suggesting that too much inclusion can be counter-productive. Indeed, a key task for future research is to further analyze and study the concept of inclusive leadership. In addition to that, scholars need to dedicate themselves to exploring the antecedents leading to successful BMI; future research could embrace other potential driving forces, both on the organization and firm levels. Thus, future research needs to explore the relationships between BMI and various environmental factors or organizational activities more systematically.

Conclusion and limitations

Based on previous research on inclusive leadership and BMI, supported by the theory of open innovation and the resource-based view, this study built a theoretical model showing that inclusive leadership has a positive influence on BMI, mediated by boundary-spanning exploration. It shows that inclusive leadership has a significant positive influence on employees' exploration and innovation in responding to the constantly changing economic environment.

Second, this study offered preliminary insights into how leaders can exercise influence through their actions and skills, and into which characteristics of inclusive leaders can promote employees' innovative behaviors, stimulate boundary-spanning exploration and facilitate BMI. On the organizational level, this study adds to the knowledge of the relationships between leadership, employees and BMI. Although scholars have identified the importance of human capital and social capital for firms' innovation (Marvel and Lumpkin, 2007; Tang and Murphy, 2012), the role of their effect on BMI has not been systematically analyzed. Thus, shifting the focus from technology to employees expands the boundaries to explore the various antecedents of BMI.

Third, this study proposed a research model that elaborates how inclusive leadership affects BMI on the organization level. These findings showed that in the attempt to pursue BMI and motivate employees' ability to innovate, boundary-spanning exploration is very important; moreover, the characteristics of openness and accessibility of inclusive leaders are more likely to guide employees to seek new knowledge, markets and products.

Limitations and suggestions for further research

First, this study only collected data and information from Zhejiang Province, China; the representativeness of the sample is, therefore, restricted. A useful extension would be to replicate the study in other emerging economies or to conduct comparative studies in both developed and developing economies. Second, the data used in this study do not allow for causal interpretations among different factors. Thus, this study cannot determine at what stage of the managerial process inclusive leadership is influential for BMI. Third, the scales used in this study were produced in the context of research carried out in Western countries and may not be as suitable for use in China; in the future, we can revise the scales to make them more suitable for use in China. China has a specific inclusive culture (Tang *et al.*, 2017)

within organizations, in which the relationship between employees and employers is subtler and more complex than in Western countries.

References

- Ahuja, G. and Lampert, C.M. (2001), "Entrepreneurship in the large corporation: a longitudinal study of how established firms create breakthrough inventions", *Strategic Management Journal*, Vol. 22 Nos 6-7, pp. 521-543.
- Amit, R. and Zott, C. (2001), "Value creation in e-business", *Strategic Management Journal*, Vol. 22 Nos 6/7, pp. 493-520.
- Amit, R. and Zott, C. (2015), "Crafting business architecture: the antecedents of business model design", *Strategic Entrepreneurship Journal*, Vol. 9 No. 4, pp. 331-350.
- Atwater, L. and Carmeli, A. (2009), "Leader-member exchange, feelings of energy and involvement in creative work", *The Leadership Quarterly*, Vol. 20, pp. 264-275.
- Baden-Fuller, C. and Haefliger, S. (2013), "Business models and technological innovation", *Long Range Planning*, Vol. 46 No. 6, pp. 419-426.
- Bandura, A. (1997), *Self-efficacy: The Exercise of Control*, W. H. Freeman, New York, NY.
- Blau, P. (1964), *Exchange and Power in Social Life*, John Wiley, New York.
- Carmeli, A., Reiter-Palmon, R. and Ziv, E. (2010), "Inclusive leadership and employee involvement in creative tasks in the workplace: the mediating role of psychological safety", *Creativity Research Journal*, Vol. 22 No. 3, pp. 250-260.
- Chesbrough, H. (2003a), *Open Innovation: The New Imperative for Creating and Profiting from Technology*, Harvard Business School Press, Boston, MA.
- Chesbrough, H. (2003b), "The era of open innovation", *MIT Sloan Management Review*, Vol. 44 No. 3, pp. 35-41.
- Chesbrough, H. (2003c), "The logic of open innovation: managing intellectual property", *California Management Review*, Vol. 45 No. 3, pp. 33-58.
- Chesbrough, H. (2010), "Business model innovation: opportunities and barriers", *Long Range Planning*, Vol. 43 Nos 2/3, pp. 354-363.
- Cohen, W.M. and Levinthal, D.A. (1990), "Absorptive capacity: a new perspective on learning and innovation", *Administrative Science Quarterly*, Vol. 35 No. 1, pp. 128-152.
- Easterbrook, M. and Vignoles, V.L. (2013), "What does it mean to belong? Interpersonal bonds and intragroup similarities as predictors of felt belonging in different types of groups", *European Journal of Social Psychology*, Vol. 43, pp. 455-462.
- Fang, Y.C. (2014), "The influence of inclusive leadership style on team performance - based on the employee's self - efficacy", *Science Research Management*, Vol. 35 No. 5, pp. 152-160.
- Fleming, L., Mingo, S. and Chen, D. (2007), "Collaborative brokerage, generative creativity, and creative success", *Administrative Science Quarterly*, Vol. 52 No. 3, pp. 443-475.
- He, H.Q., Liao, B.B. and Wen, Q.Y. (2007), "Diversity management of China labor force in the 21st century", *Journal of University of South China (Social Science Edition)*, Vol. 8, pp. 38-40, (In Chinese).
- Henderson, R. and Cockburn, I. (1994), "Measuring competence? Exploring firm effects in pharmaceutical research", *Strategic Management Journal*, Vol. 15, pp. 63-84.
- Hollander, E.P. (2009), *Inclusive Leadership: The Essential Leader-Follower Relationship*, Routledge, New York, NY.
- Hunter, S.T., Bedell, K.E. and Mumford, M.D. (2007), "Climate for creativity: a quantitative review", *Creativity Research Journal*, Vol. 19, pp. 69-90.

- Jiang, Y. and Chen, C.C. (2018), "Integrating knowledge activities for team innovation: effects of transformational leadership", *Journal of Management*, Vol. 44, pp. 1819-1847.
- Johnson, M.W., Christensen, C.C. and Kagermann, H. (2008), "Reinventing your business model", *Harvard Business Review*, Vol. 86 No. 12, pp. 50-59.
- Katz, D. and Kahn, R.L. (1978), *The Social Psychology of Organizations*, 2nd ed., Wiley, New York, NY.
- Kearney, E. and Gebert, D. (2009), "Managing diversity and enhancing team outcomes: the promise of transformational leadership", *Journal of Applied Psychology*, Vol. 94, pp. 77-89.
- Khanagha, S., Volberda, H. and Oshri, I. (2014), "Business model renewal and ambidexterity: structural alteration and strategy formation process during transition to a cloud business model", *R&D Management*, Vol. 44 No. 3, pp. 322-340.
- Kogut, B. and Zander, U. (1992), "Knowledge of the firm, combinative capabilities and the replication of technology", *Organization Science*, Vol. 3 No. 3, pp. 383-397.
- Koput, K. (1997), "A chaotic model of innovative search: some answers, many questions", *Organization Science*, Vol. 8 No. 5, pp. 528-542.
- Laursen, K. and Salter, A.J. (2006), "Open for innovation: the role of openness in explaining innovation performance among UK manufacturing firms", *Strategic Management Journal*, Vol. 27 No. 2, pp. 131-150.
- Leiponen, A. and Helfat, C.E. (2010), "Innovation objectives, knowledge source, and the benefits of breadth", *Strategic Management Journal*, Vol. 31 No. 2, pp. 224-236.
- Leiponen, A. and Helfat, C.E. (2011), "Location, decentralization, and knowledge sources for innovation", *Organization Science*, Vol. 22 No. 3, pp. 641-658.
- Leonard-Barton, D. (1995), *Wellsprings of Knowledge: Building and Sustaining the Sources of Innovation*, Harvard Business School Press, Boston, MA.
- Liu, J., Wang, H., Hui, C. and Lee, C. (2012), "Psychological ownership: how having control matters", *Journal of Management Studies*, Vol. 49, pp. 869-895.
- Martins, E.C. and Terblanche, F. (2003), "Building organizational culture that stimulates creativity and innovation", *European Journal of Innovation Management*, Vol. 6, pp. 64-74.
- Marvel, M.R. and Lumpkin, G. (2007), "Technology entrepreneur's human capital and its effects on innovation radicalness", *Entrepreneurship Theory and Practice*, Vol. 31 No. 6, pp. 807-828.
- Massa, L. and Tucci, C.L. (2013), *Business Model Innovation. Forthcoming in Oxford Handbook of Innovation Management*, Oxford University Press, Oxford.
- Metzger, J., Kraemer, N. and Terzidis, O. (2016), "A systematic approach to business modeling based on the Value Delivery Modeling Language: complexity in entrepreneurship", in Berger, E. and Kuckertz, A. (Eds), *Complexity in Entrepreneurship, Innovation and Technology Research*, Springer, Cham, pp. 245-266.
- Mezger, F. (2014), "Toward a capability-based conceptualization of business model innovation: insights from an explorative study", *R&D Management*, Vol. 44 No. 5, pp. 429-449.
- Mor Barak, M.E. (2000), "The inclusive workplace: an ecosystems approach to diversity management", *Social Work*, Vol. 45 No. 4, pp. 339-353.
- Mumford, M.D. and Hunter, S.T. (2005), "Innovation in organizations: a multi-level perspective on creativity", *Multi-Level Issues in Strategy and Methods*, Vol. 4, pp. 11-73.
- Nembhard, I.M. and Edmondson, A.C. (2006), "Making it safe: the effects of leader inclusiveness and professional status on psychological safety and improvement efforts in health care teams", *Molecular Biology of the Cell*, Vol. 27 No. 7, pp. 941-966.
- Nishii, L.H. (2013), "The benefits of climate for inclusion for gender diverse groups", *Academy of Management Journal*, Vol. 56 No. 6, pp. 1754-1774.
- Peng, S.Q., Han, J. and Zhao, Z.Y. (2008), "Employee diversity management and corporate innovation", *Management World*, Vol. 24, pp. 182-183, (In Chinese).

-
- Podsakoff, P.M., MacKenzie, S.B., Lee, J.-Y. and Podsakoff, N.P. (2003), "Common method biases in behavioral research: a critical review of the literature and recommended remedies", *Journal of Applied Psychology*, Vol. 88 No. 5, pp. 879-903.
- Reiter-Palmon, R. and Illies, J.J. (2004), "Leadership and creativity: understanding leadership from a creative problem-solving perspective", *Leadership Quarterly*, Vol. 15, pp. 55-77.
- Ruef, M. (2002), "Strong ties, weak ties and islands: structural and cultural predictors of organizational innovation", *Industrial and Corporate Change*, Vol. 11 No. 3, pp. 427-449.
- Sidhu, J.S., Commandeur, H.R. and Volberda, H.W. (2007), "The multifaceted nature of exploration and exploitation: value of supply, demand, and spatial search for innovation", *Organization Science*, Vol. 18 No. 1, pp. 20-38.
- Smith, W.K. and Tushman, M.L. (2005), "Managing strategic contradictions: a top management model for managing innovation streams", *Organization Science*, Vol. 16 No. 5, pp. 522-536.
- Snyder, C., Rand, K.R. and Sigmon, D.R. (2005), *Handbook of Positive Psychology*, Hope theory, pp. 257-276.
- Sofka, W. and Grimpe, C. (2010), "Specialized search and innovation performance — evidence across Europe", *R&D Management*, Vol. 40 No. 3, pp. 310-323.
- Stamper, C.L. and Masterson, S.S. (2002), "Insider or outsider? How employee perceptions of insider status affect their work behavior", *Journal of Organizational Behavior*, Vol. 23, pp. 875-894.
- Stuart, T.E. and Podolny, J.M. (1996), "Local search and the evolution of technological capabilities", *Strategic Management Journal*, Vol. 17 No. S1, pp. 21-38.
- Tang, J. and Murphy, P.J. (2012), "Prior knowledge and new product and service introductions by entrepreneurial firms: the mediating role of technological innovation", *Journal of Small Business Management*, Vol. 50 No. 1, pp. 41-62.
- Tang, C. and Naumann, S.E. (2016), "Team diversity, mood, and team creativity: the role of team knowledge sharing in Chinese R&D teams", *Journal of Management and Organization*, Vol. 22, pp. 420-434.
- Tang, N.Y., Zheng, X.C. and Chen, C.N. (2017), "Managing Chinese diverse workforce: toward a theory of organizational inclusion", *Nankai Business Review International*, Vol. 8 No. 1, pp. 39-56.
- Teece, D.J. (2010), "Business models, business strategy and innovation", *Long Range Planning*, Vol. 43 Nos 2/3, pp. 172-194.
- Teece, D.J., Pisano, G. and Shuen, A. (1997), "Dynamic capabilities and strategic management", *Strategic Management Journal*, Vol. 18 No. 7, pp. 509-533.
- Tyler, T.R., Degoey, P. and Smith, H. (1996), "Understanding why the justice of group procedures matters: a test of the psychological dynamics of the group-value model", *Journal of Personality and Social Psychology*, Vol. 70, pp. 913-930.
- Warner, M. and Zhu, Y. (2010), "Labour and management in the People's Republic of China: seeking the 'harmonious society'", *Asia Pacific Business Review*, Vol. 16 No. 3, pp. 285-298.

- Zhang, H. and Hu, B. (2017), "The effects of organizational isomorphism on innovation performance through knowledge search in industrial cluster", *Chinese Management Studies*, Vol. 11 No. 2, pp. 209-229.
- Zhou, J. (1998), "Feedback valence, feedback style, task autonomy, and achievement orientation: interactive effects on creative behavior", *Journal of Applied Psychology*, Vol. 83, pp. 261-276.
- Zott, C. and Amit, R. (2007), "Business model design and the performance of entrepreneurial firms", *Organization Science*, Vol. 18 No. 2, pp. 181-199.

Corresponding author

Xinxing Dai can be contacted at: 443661239@qq.com

For instructions on how to order reprints of this article, please visit our website:

www.emeraldgroupublishing.com/licensing/reprints.htm

Or contact us for further details: permissions@emeraldinsight.com